

PronounSE: SFX Synthesizer from Language-Independent Vocal Mimic Representation

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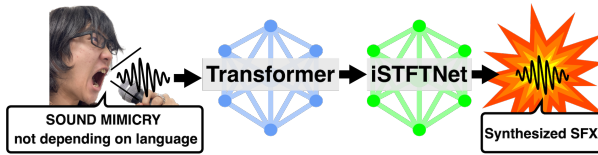


Figure 1: Schematic image of our proposed method

ABSTRACT

Sound creators make various sound effects (SFX) depending on auditory events utilizing knowledge, techniques, and experience. These are challenging tasks for inexperienced creators. In this research, we focus on the fact that it is relatively easy for anyone to mimic SFX with utterances, and we propose a novel interactive technique called PronounSE. It can synthesize SFX to reflect subtle sound nuances by the vocal representation with language-independent sound mimicry. PronounSE consists of a Transformer that converts a mel-spectrogram of utterance mimic sound into a mel-spectrogram of SFX, and iSTFTNet, as a neural-vocoder, reconstructs a waveform for a synthesized SFX. We built a dataset for PronounSE that especially picked explosion sounds which we could easily represent with many nuances. This paper describes the model of PronounSE, the dataset of explosion sounds and their various vocal mimic representations of plural people, and the results of synthesized SFXs from untrained representations interactively.

CCS CONCEPTS

• Computing methodologies → Neural networks.

KEYWORDS

Generative model, Sound Synthesis, Sound Effects, Transformer, Neural Vocoder, Explosion Sound

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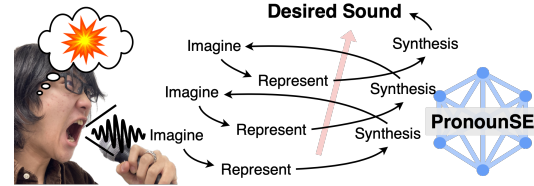


Figure 2: Sound creation flow with PronounSE

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1 INTRODUCTION

Various SFXs are utilized depending on auditory situations and events in media works, and they need to be produced. Specifically, sound creators produce them through challenging procedures for beginners, such as selecting suitable sound materials from massive databases, editing with dedicated software, and recording with the Foley technique[1][2][3]. Although recent generative AI enables everyone to create illustrations, movies, and music easily[4][5][6][7], there are few studies for SFX synthesis. As SFX techniques, except for speech and music, there are the environmental sound synthesis models from onomatopoeic words[8][9] and a non-voice sound synthesis model from text descriptions[10]. These techniques enabled a wide variety of sound synthesis; however, they still need to be improved to control subtle variations of sound nuances. These inputs are symbols of compressed sound information without the sound duration, intonation, and rhythm. On the other hand, it is possible to represent these nuances by uttering them in an SFX-mimicking manner. Then, we proposed the method to synthesize SFX from language-independent utterance mimicry voice[11], which enabled us to obtain the desired SFX through repeating speech and synthesis interactively. In this paper, we describe that our proposed model achieved significant improvements in synthesis quality, nuance follow-ability through dataset expansion, and model modifications by replacing the neural-vocoder with iSTFTNet[12] as shown in Figure 1. This paper also describes typical results in current with untrained mimic utterances for explosion sounds. In addition, we confirmed it can be used for sound creation flow shown in Figure 2.

2 SYNTHESIS SFX FROM UTTERANCE MIMICRY VOICE

2.1 Improved synthesis model

Figure 3 shows an overview of the generating process of our proposed model. It contains three parts: Pre-process, Conversion, and Waveform Reconstruction. In the Pre-Process part, a waveform of

length $512t$ is converted into a mel-spectrogram of $256 \times t$ matrices. In Conversion, Transformer[13] converts a mel-spectrogram of an utterance mimic voice into a mel-spectrogram of SFX. Specifically, a mel-spectrogram that has expanded its vertical axis, i.e., frequency resolutions from 256 to 512, is processed via encoders and decoders of 3 layers, respectively. Also, the estimated mel-spectrogram is reverted to its original dimensions. In the Waveform Reconstruction, the neural-vocoder iSTFTNet[12] estimates magnitude and phase spectrograms by up-sampling from mel-spectrograms and then synthesizes a waveform with iSTFT from an estimated spectrogram. Hence, our proposed model comprises two different neural networks trained separately.

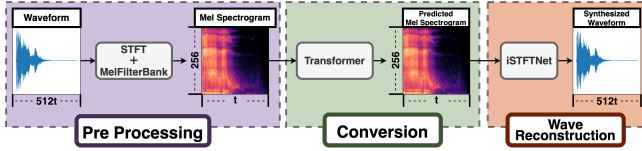


Figure 3: Process of generating sound

2.2 Expanded mimic voice dataset

To train the Transformer, the dataset comprising pairs of SFX and corresponding mimic voice is required. To date, we have collected approximately 1,300 SFXs focusing on explosion sounds containing gunshots, fires, blasts, blowups, bursts, etc., which included many nuances and were relatively easy to mimic. Our previous dataset recorded 3,575 mimic voice samples corresponding to the SFXs by only one speaker; this time, we recorded an additional 4,193 mimic voices by three speakers. Thus, we built an expanded dataset of 7,768 pairs with various voices and representations.

3 TRAINING AND SYNTHESIS

In the current PronounSE, the Transformer was trained on the dataset of pairs for 20,000 epochs with a batch size of 64. The iSTFTNet was trained on only explosion sounds for 10,000 epochs with a batch size of 64. Using these trained models, we made a web application to record voices and generate SFX sounds interactively, as shown in Figure 4.

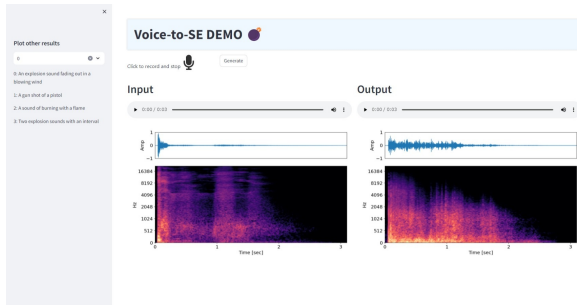


Figure 4: Web Application of PronounSE

We prepared untrained mimic voices by three speakers. Figure 5 shows two results of these untrained voices. The result 1, left, is an explosion sound fading out in a blowup. The result 2, right, is

an explosion sound with a wind blowing up strongly at the end. In both results, upper mel-spectrograms show mimic voices as inputs and lowers show generated explosion sounds. In Result 1 of Figure 5, areas surrounded by dotted green line boxes show three parts of time-series variations with the nuance represented a blowing wind. Its generated mel-spectrogram shows those represented features reflected in synthesized sounds. In Result 2 of Figure 5, a feature of a mimic voice on the right side around 2 seconds, surrounded by dotted green line boxes, shows the nuance of a wind blowing up strongly, and the feature appears in a synthesized sound at an appropriate time. These results show that our Voice-to-SFX conversion model can apply at least explosion sounds reflecting nuances of utterance mimic voices.

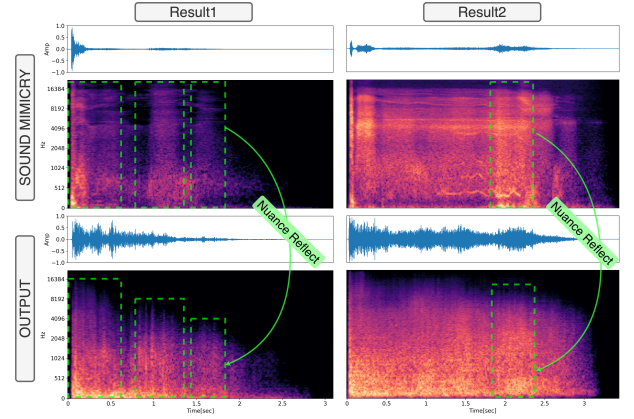


Figure 5: Results of generated explosion sounds

4 DISCUSSION

PronounSE can synthesize an SFX briefly after pronouncing a mimic sound. Thus, this can provide a novel method to create/obtain SFX materials or sound snippets for various creative people who can pronounce and represent as mimic voices of SFXs. It is not choosing it from sound libraries, not making it with synthesizer equipment, and not recording it as Foley. In addition, a creator can get a sound in a few seconds after representing it in vocals, check between it and the imagined sound in their brain, and represent it again and recheck the regenerated sound again. The repetitive sound creation can bring the sound close to their imagined sound, as shown in Figure 2. This SFX generation method from a vocal representation also has an aspect of human augmentation that could make the most of a human's oral expression ability.

5 CONCLUSION

In this study, we proposed a novel technique to synthesize SFX from the vocal mimic to enable anyone to create desired sounds easily. In this paper, we describe the expanded dataset focused on explosion sounds with three speakers, the results of using the latest trained model. It is confirmed that the model could synthesize explosion sounds corresponding to various nuance representations, such as articulation, rhythm, and intonation, and that the synthesis quality was improved. For future work, we plan to make unrealistic SFXs, for instance, laser beams or magical sounds, with building datasets.

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